

Alternative Regulators in QED

Peskin & Schroeder, An Introduction to Quantum Field Theory, Chapter 7.

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Quantum Field Theory

We have seen that the Ward identity can be violated by an improperly chosen regulator. Let us check the validity of the identity $Z_1 = Z_2$, to order α , for several choices of regulator. We have already verified that the relation holds for Pauli-Villars regularisation.

(a) First, let us verify the identity, $\delta Z_1 = \delta Z_2$, by simply replacing introducing an UV cutoff, Λ , on the integration over l_E . Recall

$$\Gamma^\mu(q=0) = Z_1^{-1} \gamma^\mu \implies \delta Z_1 = -\delta F_1(q^2=0), \quad (1)$$

and

$$Z_2^{-1} = 1 - \frac{d\Sigma_2}{d\cancel{p}} \Big|_{\cancel{p}=m} \implies \delta Z_2 = \frac{d\Sigma_2}{d\cancel{p}} \Big|_{\cancel{p}=m_0}. \quad (2)$$

(We have used the fact that $m = m_0$ at tree-level.)

Let us start by evaluating δZ_2 . From Peskin (7.17), we have

$$\Sigma_2(p) = -ie^2 \int_0^1 dx \int \frac{d^4 l}{(2\pi)^4} \frac{-2x\cancel{p} + 4m_0}{(l^2 - \Delta + i\varepsilon)^2}, \quad (3)$$

where $\Delta = -x(1-x)p^2 + (1-x)m_0^2$. Perform a Wick rotation on the integral, and we obtain

$$\begin{aligned} \int \frac{d^4 l}{(2\pi)^4} \frac{1}{(l^2 - \Delta + i\varepsilon)^2} &= i \int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{(l_E^2 + \Delta)^2} \\ &= \frac{i}{8\pi^2} \int_0^\infty dl_E \frac{l_E^3}{(l_E^2 + \Delta)^2}. \end{aligned} \quad (4)$$

Now, we can apply the UV cutoff to the integral, which yields

$$\frac{i}{8\pi^2} \int_0^\Lambda dl_E \frac{l_E^3}{(l_E^2 + \Delta)^2} = \frac{i}{16\pi^2} \left[\ln \left(1 + \frac{\Lambda^2}{\Delta} \right) - \frac{\Lambda^2}{\Lambda^2 + \Delta} \right]. \quad (5)$$

Thus, by differentiating Σ_2 and noting that $p^2 = \cancel{p}^2$, we can obtain

$$\begin{aligned} \delta Z_2 &= -\frac{e^2}{8\pi^2} \int_0^1 dx \frac{\partial}{\partial \cancel{p}} \left\{ (x\cancel{p} - 2m_0) \left[\ln \left(1 + \frac{\Lambda^2}{\Delta} \right) - \frac{\Lambda^2}{\Lambda^2 + \Delta} \right] \right\} \Big|_{\cancel{p}=m_0} \\ &= -\frac{\alpha}{2\pi} \int_0^1 dx \left\{ x \left[\ln \left(1 + \frac{\Lambda^2}{\Delta} \right) - \frac{\Lambda^2}{\Lambda^2 + \Delta} \right] - 2(x\cancel{p} - 2m_0)\cancel{p}x(1-x) \left[-\frac{\Lambda^2}{\Delta(\Lambda^2 + \Delta)} + \frac{\Lambda^2}{(\Lambda^2 + \Delta)^2} \right] \right\} \Big|_{\cancel{p}=m_0} \\ &= -\frac{\alpha}{2\pi} \int_0^1 dx \left\{ x \left[\ln \left(1 + \frac{\Lambda^2}{\Delta} \right) - \frac{\Lambda^2}{\Lambda^2 + \Delta} \right] + 2m_0^2(x-2)x(1-x) \frac{\Lambda^4}{\Delta(\Lambda^2 + \Delta)^2} \right\} \Big|_{\Delta=(1-x)^2 m_0^2} \\ &= -\frac{\alpha}{2\pi} \int_0^1 dx \left\{ x \ln \left(1 + \frac{\Lambda^2}{m_0^2(1-x)^2} \right) - x + \frac{2(x-2)x}{(1-x)} + \mathcal{O}\left(\frac{m_0^2}{\Lambda^2}\right) \right\}. \end{aligned} \quad (6)$$

Now, let us turn to δZ_1 . From Peskin (6.54), we have

$$\delta F_1(q^2=0) = 2ie^2 \int_0^1 dx dy dz \delta(x+y+z-1) \int \frac{d^4 l}{(2\pi)^4} \frac{-l^2 + 2(1-4x+x^2)m_0^2}{(l^2 - \Delta + i\varepsilon)^3}, \quad (7)$$

where $\Delta = (1-x)^2 m_0^2$. Again, Wick rotate the integral with respect to l , we obtain

$$\begin{aligned} \int \frac{d^4 l}{(2\pi)^4} \frac{-l^2 + 2(1-4x+x^2)m_0^2}{(l^2 - \Delta + i\varepsilon)^3} &= -i \int \frac{d^4 l_E}{(2\pi)^4} \frac{l_E^2 + 2(1-4x+x^2)m_0^2}{(l_E^2 + \Delta)^3} \\ &= \frac{-i}{8\pi^2} \int_0^\infty dl_E l_E^3 \cdot \frac{l_E^2 + 2(1-4x+x^2)m_0^2}{(l_E^2 + \Delta)^3}. \end{aligned} \quad (8)$$

Let us regulate the integral with a UV cutoff:

$$\int_0^\infty \rightarrow \int_0^\Lambda. \quad (9)$$

Thus,

$$\begin{aligned}
& \frac{-i}{8\pi^2} \int_0^\Lambda dl_E l_E^3 \cdot \frac{l_E^2 + 2(1-4x+x^2)m_0^2}{(l_E^2 + \Delta)^3} = \frac{-i}{8\pi^2} \int_{\sqrt{\Delta}}^{\sqrt{\Lambda^2 + \Delta}} dk \left[\frac{(k^2 - \Delta)^2}{k^5} + 2(1-4x+x^2)m_0^2 \frac{k^2 - \Delta}{k^5} \right] \\
&= \frac{-i}{8\pi^2} \left[\frac{1}{2} \ln \left(1 + \frac{\Lambda^2}{\Delta} \right) + \frac{\Delta}{\Lambda^2 + \Delta} - \frac{3}{4} - \frac{\Delta^2}{4(\Lambda^2 + \Delta)^2} + 2(1-4x+x^2)m_0^2 \left(-\frac{1}{2(\Lambda^2 + \Delta)} + \frac{1}{4\Delta} + \frac{\Delta}{4(\Lambda^2 + \Delta)} \right) \right] \\
&= \frac{-i}{8\pi^2} \left[\frac{1}{2} \ln \left(1 + \frac{\Lambda^2}{\Delta} \right) - \frac{3}{4} + \frac{(1-4x+x^2)m_0^2}{2\Delta} + \mathcal{O}\left(\frac{m_0^2}{\Lambda^2}\right) \right].
\end{aligned} \tag{10}$$

Therefore, we have

$$\delta Z_1 = -\frac{\alpha}{2\pi} \int_0^1 dx \int_0^{1-x} dy \left[\ln \left(1 + \frac{\Lambda^2}{\Delta} \right) - 1 + \mathcal{O}\left(\frac{m_0^2}{\Lambda^2}\right) \right]. \tag{11}$$

Hence, we have shown that $\delta Z_1 \neq \delta Z_2$ under the UV cutoff regulator.

(b) In this section, let us see what happened with dimensional regularisation. Note that in d dimensions,

$$\gamma^\mu \gamma^\nu \gamma_\mu = -(2 - \epsilon) \gamma^\nu; \tag{12}$$

$$\gamma^\mu \gamma_\mu = d. \tag{13}$$

Then, Peskin (7.17) is modified to be

$$\begin{aligned}
\Sigma_2(p) &= ie^2 \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{(2-d)x\not{p} + d \cdot m_0}{(l^2 - \Delta + i\epsilon)^2} \\
&= -e^2 \int_0^1 dx \int \frac{d^d l_E}{(2\pi)^d} \frac{(2-d)x\not{p} + d \cdot m_0}{(l_E^2 + \Delta)^2} \\
&= \frac{-e^2}{(4\pi)^{d/2}} \Gamma(2 - \frac{d}{2}) \int_0^1 dx [(2-d)x\not{p} + d \cdot m_0] \cdot \frac{1}{\Delta^{2-d/2}}.
\end{aligned} \tag{14}$$

Thus,

$$\delta Z_2 = \frac{d\Sigma_2}{d\not{p}} \Big|_{\not{p}=m_0} = -\frac{2e^2}{(4\pi)^{d/2}} \Gamma(2 - \frac{d}{2}) \int_0^1 dx \left[\frac{(2-d)x}{\Delta^{2-d/2}} + (2 - \frac{d}{2}) \frac{[(2-d)xm_0 + d \cdot m_0]2x(1-x)m_0}{\Delta^{3-d/2}} \right] \Big|_{\Delta=(1-x)^2 m_0^2}. \tag{15}$$

Now, take the limit $d = 4 - \epsilon$, where $\epsilon \rightarrow 0$, we have

$$\begin{aligned}
\delta Z_2 &= -\frac{e^2}{(4\pi)^2} \left(1 + \frac{\epsilon}{2} \ln 4\pi \right) \left(\frac{2}{\epsilon} - \gamma + \mathcal{O}(\epsilon) \right) \int_0^1 dx \\
&\quad \times \left[-2x + \epsilon[(\epsilon-2)xm_0 + (4-\epsilon)m_0]2x(1-x)m_0 \cdot \frac{1}{\Delta} \right] \left(1 - \frac{1}{2}\epsilon \ln \Delta + \mathcal{O}(\epsilon^2) \right) \\
&= \frac{\alpha}{2\pi} \int_0^1 dx (2x) \left[\frac{2}{\epsilon} - \gamma - \ln 4\pi - 1 - \ln m_0^2 - 2 \ln(1-x) - \frac{2(2-x)}{1-x} \right] + \mathcal{O}(\epsilon).
\end{aligned} \tag{16}$$